

Accuracy-interpretability tradeoff: compact symbolic formulas vs gradient-boosted ensembles

Acidic pK_a

GBM ensembles
 $R^2=0.78-0.84$
(1133 descriptors)

QLattice
★ 8 feats, $R^2=0.45$

distilled

Basic pK_a

GBM ensembles
 $R^2=0.80-0.84$
(1707 descriptors)

QLattice
★ 12 feats, $R^2=0.47$

distilled

Held-out test R^2

Number of input descriptors (log scale)

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